

cft-anyons Lab Book
From category data to provably rigorous CFTs

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1 Frontmatter, North Star, and Claim Status

This is a reproducible research lab book. Its guiding goal — the *north star* — is a single, concrete construction problem:

Given a (unitary) fusion or modular tensor category $\mathcal{C}$, together with whatever additional data is needed (OPE coefficients / conformal data), construct a family of microscopic (lattice) models and their symmetry generators whose continuum limit *provably* yields a mathematically rigorous QFT/CFT that (i) realises the input category data and (ii) carries a full projective unitary representation of the symmetries.

The working belief is that this is achievable at least for *rational* CFTs, and perhaps for all QFTs. This is a programme, not a settled theory: every step toward the goal is a question, proposal, or hypothesis until a local source, a local derivation, or a reproducible run supports it.

1.1 The Two Success Criteria

A construction is only a candidate solution if it meets both, non-negotiably:

1. **Provability.** The continuum limit is established rigorously (an operator-algebraic / wavelet renormalisation limit with theorems, not a heuristic scaling argument).
2. **Fidelity.** The realised continuum content faithfully reproduces the input category data $\mathcal{C}$, and the symmetry representation is *full* and projective-unitary.

1.2 Claim Status

No claim is treated as established until it is marked *Source-local* or *Checked* and linked to a local source, derivation, test, or run artifact. The status labels are fixed in `CONVENTIONS.md`: *Question*, *Proposal*, *Source-local*, *Checked*, and *Rejected*. These labels are part of the scientific record; they separate programme-level conjecture from results.

1.3 Evidence Chain

Every substantive assertion should be traceable through:

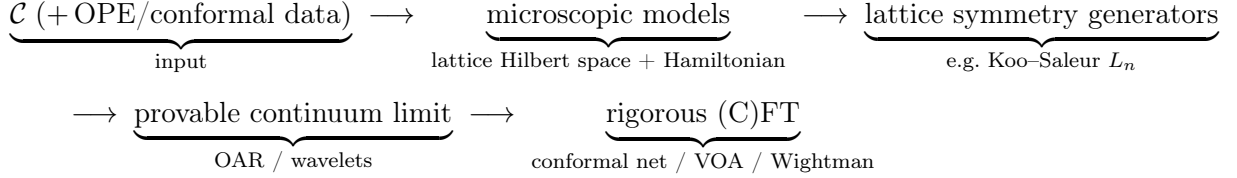
1. a local source manifest under `references/` (the kept papers are the ground truth),
2. a convention entry in `CONVENTIONS.md`,
3. a derivation, script, test, or run bundle when computation is involved,
4. an index row in `INDEX.md`, and
5. a report shard in this lab book.

Missing links are allowed during exploration only when they are explicit.

2 Programme Map

This shard is a map of questions, not a list of results. The north star (§1 / CA-00) factors into a pipeline of stages; each stage is an open problem with its own conventions, sources, and checkable invariants.

2.1 The Pipeline



2.2 Stage 1 — Category Input

- Which categories are in initial scope: multiplicity-free unitary fusion categories (Fibonacci, Ising/TL), then modular tensor categories?
- What auxiliary data beyond $\mathcal{C}$ is genuinely required (OPE coefficients, central charge, conformal weights), and for which target CFTs?
- Which gauge and indexing conventions are fixed before any F/R-symbol is written (see CONVENTIONS.md)?

2.3 Stage 2 — Microscopic Models

- Which family realises $\mathcal{C}$ on the lattice: anyon chains, string-nets, or another construction — and how is the family indexed by scale?
- What is the lattice Hilbert space (fusion-tree basis) and the Hamiltonian, and which invariants pin them (fusion-rule consistency, pentagon, projector idempotence)?

2.4 Stage 3 — Lattice Symmetry Generators

- Which lattice operators approximate the continuum symmetry generators (Koo-Saleur lattice Virasoro L_n , $\bar{L}_n$, and kin)?
- What convergence statement is the target, and what is checkable at finite size (commutator algebra, central charge extraction)?

2.5 Stage 4 — Provable Continuum Limit

- Which rigorous renormalisation framework carries the limit (operator- algebraic renormalisation, wavelet / multi-scale entanglement)?
- What must be proved: existence of the limit, the limiting Hilbert space, and convergence of the symmetry generators to a projective unitary representation?

2.6 Stage 5 — Rigorous (C)FT Target

- What is the target object: a conformal net, a vertex operator algebra, or a Wightman/Osterwalder–Schrader theory?
- What is the fidelity statement: that the target’s superselection / representation category recovers $\mathcal{C}$?

2.7 First Decisions

The first technical session should not start with formulas. It should register the sources for one worked input category (Fibonacci or Ising/TL), fix the minimum conventions (F/R-symbol gauge, vacuum index, fusion-tree ordering), and identify the smallest invariant that can be checked end-to-end on one stage of the pipeline.

3 Many-Particle Hilbert Spaces: the AND/OR/Vacuum Grammar

3.1 Sources and Dependencies

This is a pre-topic foundation for the programme (see CA-01, the programme map): how to build the Hilbert space of an indefinite number of particles of arbitrary types. The guiding idea is that two logical connectives map cleanly onto two Hilbert-space constructions, and the no-particle system is the complex line.

This shard treats the *finite* grammar and its semantics; the rigorous *indefinite*-particle (Hilbert-completed) case is CA-03. It harmonises with, and re-proves nothing from, the finite “AND/OR many-particle grammar” developed by the author in the sibling project (notation $\mathcal{V}$, $\mathcal{V}_n$, AND = $\otimes$, OR = $\oplus$, $\mathbf{1} = \mathbb{C}$); see that project’s shard AQM-70. Statistics (Bose / Fermi / para) are deferred throughout.

3.2 The Three Primitives

Let particle *types* be (separable, complex) Hilbert spaces; a type’s space is the state space of one particle of that type. Three primitives generate composite systems:

- **OR = direct sum** $\oplus$. “Either a particle of type 1 or of type 2” is $\mathcal{H}_1 \oplus \mathcal{H}_2$.
- **AND = tensor product** $\otimes$. “A particle of type 1 *and* a particle of type 2” is $\mathcal{H}_1 \otimes \mathcal{H}_2$.
- **vacuum** = $\mathbb{C}$. The no-particle system is the complex line.

Crucially, this grammar fixes *indefinite particle number*; exchange symmetry (statistics) is a *separate, later* step, realised as a quotient or subspace (CA-03, §“Symmetries come later”). Keeping the two apart is the whole point.

3.3 Two Monoidal Structures and Their Units

On the category Hilb of complex Hilbert spaces (with the obvious unitary coherence isomorphisms) there are two symmetric monoidal structures:

$$(\text{Hilb}, \oplus, \{0\}) \quad \text{and} \quad (\text{Hilb}, \otimes, \mathbb{C}),$$

with the two units *distinct*:

$$A \oplus \{0\} \cong A \quad (\oplus\text{-unit is the zero space } \{0\}), \quad A \otimes \mathbb{C} \cong A \quad (\otimes\text{-unit is the vacuum } \mathbb{C}).$$

The two are linked by a *distributivity* natural isomorphism, which on Hilbert spaces is *unitary*:

$$\delta : A \otimes (B \oplus C) \xrightarrow{\cong} (A \otimes B) \oplus (A \otimes C), \quad A \otimes \{0\} \cong \{0\}.$$

Together $(\text{Hilb}, \oplus, \{0\}, \otimes, \mathbb{C})$ is a symmetric *rig* (bimonoidal) category. Its coherence — that all diagrams built from associators, unitors, symmetries, and δ commute — is standard (Laplaza; Kelly) and is *not* re-proved here; downstream we use only the elementary, manifestly unitary associativity/unitor isomorphisms and the degree-0 identification $\mathbb{C} \otimes A \cong A$.

Vacuum = $\mathbb{C}$, **not** $\{0\}$. The vacuum is the $\otimes$ -unit $\mathbb{C}$ (the empty tensor product, one particle’s worth of “nothing”), a *one-dimensional* space carrying the unit vector Ω . It is *not* the $\oplus$ -unit $\{0\}$ (the empty direct sum, the impossible system). Empty $\otimes$ is $\mathbb{C}$; empty $\oplus$ is $\{0\}$. Conflating them is the first pitfall this grammar guards against (cf. CONVENTIONS.md (a), which fixes the vacuum *object* as $\mathbb{C}$).

3.4 The Grammar and its Carrier

Fix a set $\{K_x\}_{x \in X}$ of single-particle type spaces (the *atoms*). Grammar expressions are

$$E ::= 0 \mid \mathbf{1} \mid K \mid \text{AND}(E, F) \mid \text{OR}(E, F),$$

where 0 is the zero atom, $\mathbf{1}$ is the vacuum atom ($\mathbf{1} = \mathbb{C}$), and K ranges over the atoms. The *carrier* $\mathcal{V}(E)$ is defined recursively:

$$\begin{aligned} \mathcal{V}(0) &= \{0\}, & \mathcal{V}(\mathbf{1}) &= \mathbb{C}, & \mathcal{V}(K) &= K, \\ \mathcal{V}(\text{AND}(E, F)) &= \mathcal{V}(E) \otimes \mathcal{V}(F), & \mathcal{V}(\text{OR}(E, F)) &= \mathcal{V}(E) \oplus \mathcal{V}(F). \end{aligned}$$

Thus AND is independent coexistence and OR is alternatives. Parenthesisation and order are part of the written expression; the rig coherence above guarantees that re-bracketing or re-ordering a *fixed* expression changes $\mathcal{V}(E)$ only by a canonical unitary isomorphism, so $\mathcal{V}$ is well defined up to canonical unitary isomorphism. (Morally $\mathcal{V}$ is the evaluation of the free symmetric rig category on $\{K_x\}$ into Hilb; we use this only as motivation, not as a load-bearing theorem.)

3.5 Particle-Number Grading

Declare each atom K to be *one* particle and $\mathbf{1}$ to be *zero* particles. The carrier then carries a particle-number grading $\mathcal{V}_n$:

$$\begin{aligned} \mathcal{V}_0(\mathbf{1}) &= \mathbb{C}, \quad \mathcal{V}_n(\mathbf{1}) = \{0\} \ (n \neq 0), & \mathcal{V}_1(K) &= K, \quad \mathcal{V}_n(K) = \{0\} \ (n \neq 1), & \mathcal{V}_n(0) &= \{0\}, \\ \mathcal{V}_n(\text{OR}(E, F)) &= \mathcal{V}_n(E) \oplus \mathcal{V}_n(F), & \mathcal{V}_n(\text{AND}(E, F)) &= \bigoplus_{i+j=n} \mathcal{V}_i(E) \otimes \mathcal{V}_j(F). \end{aligned}$$

For a *finite* expression the parse tree is finite, so only finitely many sectors are nonzero and $\mathcal{V}(E) = \bigoplus_{n \geq 0} \mathcal{V}_n(E)$ is a *finite* direct sum. The integer n is a grammar count of one-particle atoms; it is not a dynamics, a statistics rule, or (yet) an indefinite-particle space.

Worked atoms. A one-particle “menu” over a finite type set X is $P = \text{OR}_{x \in X} K_x$, with $\mathcal{V}(P) = \bigoplus_{x \in X} K_x$ (the infinite, ℓ^2 -completed menu is CA-03). The expression $\mathbf{1} \text{ OR } K$ has carrier $\mathbb{C} \oplus K$ — a “maybe” particle (either zero or one particle of type K), as in the sibling shard’s $\mathbb{C} \oplus \mathbb{C}^d$. Tensoring N “maybe” atoms, $\text{AND}_{j=1}^N (\mathbf{1} \text{ OR } K)$, has carrier $(\mathbb{C} \oplus K)^{\otimes N}$, whose grade- n sector (by distributivity) is $\bigoplus_{|S|=n} \bigotimes_{x \in S} K$ over n -subsets of the N slots — the finite “occupation” picture.

3.6 Lamport Dependency Skeleton

- D1 : atoms $\{K_x\}_{x \in X}$ (single-particle Hilbert spaces); $\mathbf{1} = \mathbb{C}$ (vacuum), $0 = \{0\}$.
- D2 : $E ::= 0 \mid \mathbf{1} \mid K \mid \text{AND}(E, F) \mid \text{OR}(E, F)$.
- D3 : $\mathcal{V} : \mathcal{V}(0) = \{0\}, \mathcal{V}(\mathbf{1}) = \mathbb{C}, \mathcal{V}(\text{AND}) = \otimes, \mathcal{V}(\text{OR}) = \oplus$.
- D4 : $\mathcal{V}_0(\mathbf{1}) = \mathbb{C}, \mathcal{V}_1(K) = K; \mathcal{V}_n(\text{OR}) = \oplus, \mathcal{V}_n(\text{AND}) = \bigoplus_{i+j=n} \mathcal{V}_i \otimes \mathcal{V}_j$.
- D5 : (Hilb, $\oplus, \{0\}, \otimes, \mathbb{C}$) a symmetric rig category, with
 - $\delta : A \otimes (B \oplus C) \xrightarrow{\cong} (A \otimes B) \oplus (A \otimes C)$ unitary and natural.
- T1 : (D5) coherence (Laplaza; Kelly) $\Rightarrow \mathcal{V}$ well defined up to canonical unitary iso.
- T2 : (D4) finite $E : \mathcal{V}(E) = \bigoplus_{n \geq 0} \mathcal{V}_n(E)$ a finite sum (finite particle number).

3.7 Scope and Deferral

Two extensions are deliberately out of this shard:

- **Indefinite particle number.** A genuine superposition over *unboundedly* many particle numbers has infinite support and does not live in the finite direct sum above; it requires the Hilbert (ℓ^2) completion. That is the content of CA-03, and is the new rigorous part of this pre-topic.
- **Statistics.** AND and OR choose no exchange symmetry. Bosonic / fermionic / para sectors arise later, by acting with the permutation group on tensor slots and passing to an invariant subspace or quotient (CA-03). This grammar only sets up the carrier those symmetries act on.

Pitfalls. (i) The $\oplus$ -unit is $\{0\}$; the $\otimes$ -unit (the vacuum) is $\mathbb{C}$ — distinct objects. (ii) Distributivity is an *isomorphism*, not an equality; all sector identities below hold up to canonical unitary iso. (iii) Everything here is *finite*; do not silently read the grading as an indefinite-particle space.

4 Indefinite Particle Number: the Full Fock Completion

4.1 Sources and Dependencies

This shard completes CA-02’s finite AND/OR/vacuum grammar to a genuine *indefinite*-particle Hilbert space. It depends on CA-02 (the two monoidal structures, the units $\{0\}$ and $\mathbb{C}$, the carrier $\mathcal{V}$ and grading $\mathcal{V}_n$). The named Fock construction is anchored locally:

The sibling project’s finite grammar (AQM-70) is recovered as the finite-truncation of the object built here.

4.2 Two Direct Sums — the Load-Bearing Distinction

Let $\{W_n\}_{n \geq 0}$ be a countable family of Hilbert spaces.

- The **algebraic direct sum** $\bigoplus_n^{\text{alg}} W_n$ consists of finite-support sequences (w_n) ($w_n \in W_n$, $w_n = 0$ for all but finitely many n). It is a pre-Hilbert space, generally *incomplete*.
- The **Hilbert (ℓ^2) direct sum** $\widehat{\bigoplus}_n W_n$ consists of sequences with $\sum_n \|w_n\|_{W_n}^2 < \infty$, with inner product $\langle (w_n), (v_n) \rangle = \sum_n \langle w_n, v_n \rangle_{W_n}$ (absolutely convergent by Cauchy–Schwarz). It is *complete*, hence a Hilbert space.

$\bigoplus_n^{\text{alg}} W_n$ is a *dense* subspace of $\widehat{\bigoplus}_n W_n$, and the latter is its completion (standard functional analysis).

Why indefinite particle number forces $\widehat{\bigoplus}$. A state with nonzero amplitude in *unboundedly* many particle-number sectors (a genuine superposition over indefinitely many particles) has infinite support: it lies in $\widehat{\bigoplus}$ but not in $\bigoplus^{\text{alg}}$. “Indefinite particle number” $\Leftrightarrow$ “not confined to finitely many sectors” $\Rightarrow$ the ℓ^2 completion is *required*. This is exactly what CA-02’s finite grammar lacks.

4.3 Completed Tensor Powers

For Hilbert spaces H, K , the *completed* (Hilbert) tensor product $H \widehat{\otimes} K$ is the completion of the algebraic tensor product under the inner product determined by $\langle a \otimes b, c \otimes d \rangle = \langle a, c \rangle_H \langle b, d \rangle_K$ (well defined and positive definite). For finite-dimensional factors $\widehat{\otimes} = \otimes$; the completion bites only for infinite-dimensional one-particle spaces. Define the *completed tensor powers*

$$H^{\widehat{\otimes} 0} := \mathbb{C} \text{ (the vacuum sector),} \quad H^{\widehat{\otimes} n} := \underbrace{H \widehat{\otimes} \cdots \widehat{\otimes} H}_n,$$

the bracketing immaterial up to the canonical unitary associativity isomorphism.

4.4 The Full Fock Space

Definition. For a one-particle Hilbert space H , the *full Fock space* is

$$\mathfrak{F}(H) := \widehat{\bigoplus_{n \geq 0} H^{\widehat{\otimes} n}}, \quad H^{\widehat{\otimes} 0} = \mathbb{C},$$

i.e. $\psi = (\psi_n)_{n \geq 0}$ with $\psi_n \in H^{\widehat{\otimes} n}$ and $\|\psi\|^2 = \sum_n \|\psi_n\|^2 < \infty$. The unit vector $\Omega = (1, 0, 0, \dots)$ in the degree-0 sector $\mathbb{C}$ is the *vacuum*. This is the *distinguishable* Fock space (the author’s term in the local source 1910.10619); in operator algebra it is also called the *free* or *Boltzmann* Fock

space (standard terminology, not anchored in a local source), namely the Hilbert completion of the tensor algebra $T(H) = \bigoplus_n H^{\otimes n}$. It is *not* the symmetric (bosonic) or antisymmetric (fermionic) Fock space; see “Symmetries come later” below.

Theorem (Fock is a Hilbert space). $\mathfrak{F}(H)$ is a Hilbert space; (i) the algebraic direct sum $\bigoplus_n^{\text{alg}} H^{\widehat{\otimes} n}$ (the finite-particle vectors) is *dense*; (ii) distinct- n sectors are *mutually orthogonal*, $\langle \psi, \varphi \rangle = \sum_n \langle \psi_n, \varphi_n \rangle$, so $H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n}$ for $m \neq n$; (iii) on the degree-0 sector the inner product is the standard one on $\mathbb{C}$ and Ω is a unit vector. *Proof.* An instance of the ℓ^2 -direct-sum facts above with $W_n = H^{\widehat{\otimes} n}$: completeness, density of the algebraic sum, and summand orthogonality are exactly those statements. $\square$

The $1/N!$ caveat (a convention trap). Schottenloher’s inner product on $D = \bigoplus_N H_N$ weights sector N by $1/N!$; that factor is a *symmetric*-Fock normalisation correcting the $N!$ overcounting of symmetric tensors. The full Fock space above uses the *plain* ℓ^2 sum with *no* combinatorial weight. Do not copy the $1/N!$ into the distinguishable case.

4.5 The Grammar Generates the Indefinite-Particle Space

Let the one-particle menu over a countable type set X be $P = \text{OR}_{x \in X} K_x$, with carrier $\mathcal{V}(P) = \widehat{\bigoplus_{x \in X} K_x}$ (the ℓ^2 direct sum; an ordinary finite direct sum when X is finite). The indefinite-particle space over this menu is $\mathfrak{F}(\mathcal{V}(P))$. Expanding each completed tensor power by distributivity (CA-02’s δ , extended unitarily to the completions) gives the $(n, \text{type-word})$ -graded decomposition

$$\mathcal{V}(P)^{\widehat{\otimes} n} \cong \widehat{\bigoplus_{w \in X^n} K_w}, \quad K_w := K_{x_1} \widehat{\otimes} \cdots \widehat{\otimes} K_{x_n}, \quad \implies \quad \mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus_{n \geq 0}} \widehat{\bigoplus_{w \in X^n} K_w}.$$

Each of the n tensor slots independently chooses a type $x_i \in X$ (two slots may choose the same type; nothing is symmetrised). The grading is $\mathcal{V}_n = \widehat{\bigoplus_{w \in X^n} K_w}$ (n = particle number, $w \in X^n$ = ordered type-word) — the ℓ^2 -completed, all- n analogue of CA-02’s finite grade- n sector. The sibling project’s finite expressions ($P_X^{\text{AND } m}$, the truncation $T_{\leq M}$) are precisely the finite-support / finite- M restrictions of this object; that is the exact harmonisation: AQM-70 is the $\bigoplus^{\text{alg}}$ /finite- M truncation of $\mathfrak{F}$.

4.6 Symmetries Come Later (the Deferred Quotient)

The full Fock space is the carrier on which exchange symmetry acts; it is not itself a (anti)symmetric space. For each n the permutation group S_n acts unitarily on $H^{\widehat{\otimes} n}$ by permuting tensor slots. Bosonic and fermionic Fock spaces are then the images of the per-sector orthogonal projectors

$$\Gamma_s(H) = P_{\text{sym}} \mathfrak{F}(H) \quad (\text{symmetric / Bose}), \quad \Gamma_a(H) = P_{\text{anti}} \mathfrak{F}(H) \quad (\text{antisymmetric / Fermi}),$$

with para-statistics the other Young-symmetry isotypic components. This is the “impose an equivalence relation under particle exchange” step. We define no projector and prove nothing here: the construction of these symmetry quotients/subspaces, and the genuinely anyonic case (where standard Fock space is not the right object), are deferred to later shards.

4.7 Lamport Dependency Skeleton

- $D6 : \bigoplus^{\text{alg}}(\text{finite support}) \subset \widehat{\bigoplus}(\ell^2) : \langle (w_n), (v_n) \rangle = \sum_n \langle w_n, v_n \rangle.$
 $D7 : \text{completed tensor power } H^{\widehat{\otimes} n}, H^{\widehat{\otimes} 0} = \mathbb{C} \text{ (vacuum).}$
 $D8 : \mathfrak{F}(H) := \widehat{\bigoplus}_{n \geq 0} H^{\widehat{\otimes} n} \text{ (full / distinguishable Fock); } \Omega = (1, 0, \dots).$
 $D9 : P = \text{OR}_{x \in X} K_x; \quad \mathfrak{F}(\mathcal{V}(P)) = \widehat{\bigoplus}_{n \geq 0} \mathcal{V}(P)^{\widehat{\otimes} n}.$
 $T3 : (D6, D7) \mathfrak{F}(H) \text{ Hilbert; } \bigoplus^{\text{alg}} \text{ dense; } H^{\widehat{\otimes} m} \perp H^{\widehat{\otimes} n} \text{ (} m \neq n \text{).}$
 $T4 : \text{indefinite particle no.} \Leftrightarrow \text{infinite support} \Rightarrow \widehat{\bigoplus} \text{ (not } \bigoplus^{\text{alg}} \text{).}$
 $T5 : (D8, \delta) \mathfrak{F}(\mathcal{V}(P)) \cong \widehat{\bigoplus}_{n \geq 0} \widehat{\bigoplus}_{w \in X^n} K_w \text{ ((} n, \text{type-word) grading).}$
 $T6 : \Gamma_s = P_{\text{sym}} \mathfrak{F}, \Gamma_a = P_{\text{anti}} \mathfrak{F} \text{ deferred (statistics later).}$

4.8 Scope and Pitfalls

In scope: the indefinite-particle direction, i.e. countable OR (the Fock completion). **Out of scope:** countable AND (an *infinite tensor product*), which is analytically a different object (von Neumann's infinite tensor product, needing a stabilising reference sequence) — not developed here.

Pitfalls: (i) *algebraic* $\bigoplus^{\text{alg}}$ (finite particle number, dense, incomplete) vs. *completed* $\widehat{\bigoplus}$ (indefinite, complete) — the distinction this shard exists to make. (ii) For infinite-dimensional one-particle H , the AND of CA-02 must mean the *completed* $\widehat{\otimes}$, not the algebraic $\otimes$. (iii) The $1/N!$ symmetric-Fock weight must not be imported into the distinguishable case. (iv) All sector identities hold on dense subspaces and extend by continuity; states are ℓ^2 -summable and cross-sector inner products vanish.